



The best defense is a good offense

A pragmatic path to
Continuous Purple Teaming

Cas van Cooten
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Who am I?

- Long-time offensive security enthusiast & Red Teamer
- Conference speaker on MalDev and building security tools
- Former: External Red Team @ Deloitte → internal Red Team @ ABN AMRO
- Current: Co-founder of **Offensys**:
a Continuous Purple Teaming platform
(but don't worry – this is not a commercial talk)



Cas van Cooten

Offensive Security Specialist

What is Purple Teaming?



THIS TALK



Red Team

The Offense

Emulates adversary TTPs to expose exploitable gaps across people, process, and technology.



Purple Team

The Collaboration

Offense and Defense collaborating in full transparency: red runs simulations and provides offensive insight, while blue hunts and improves control posture.

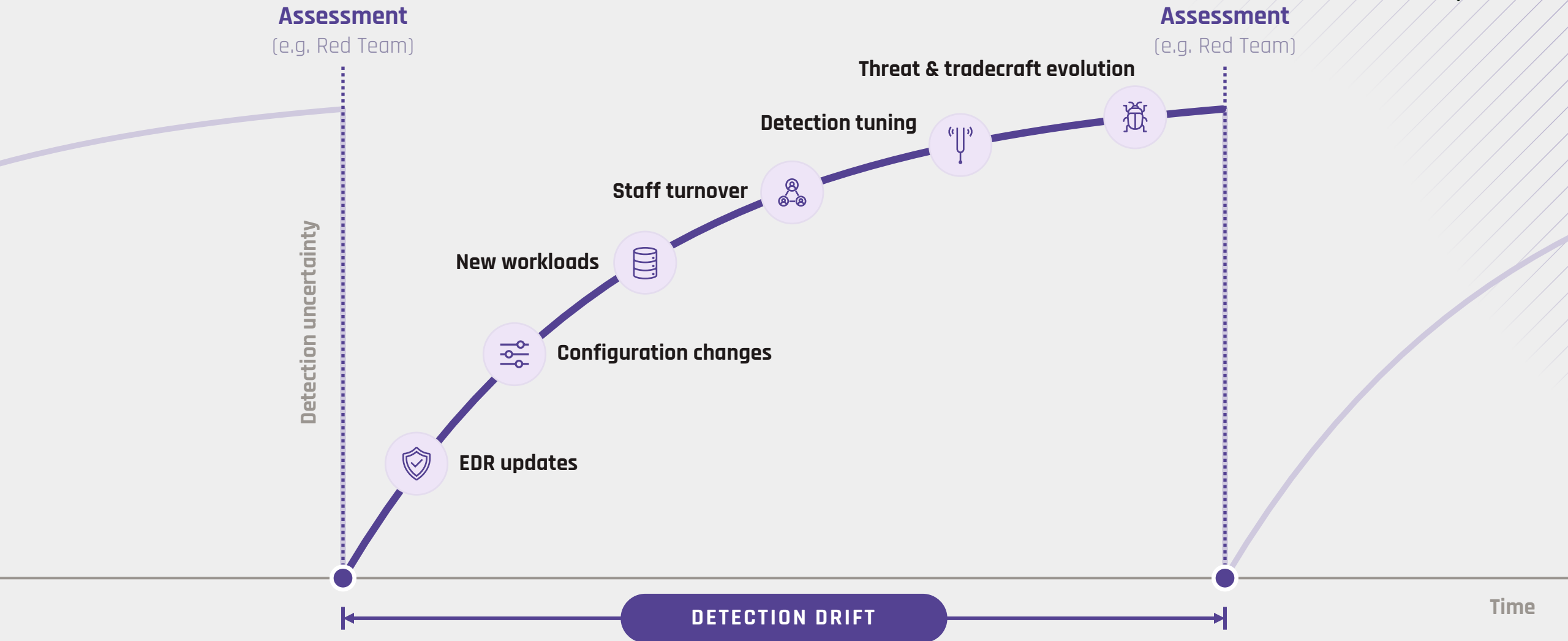


Blue Team

The Defense

Prevents, detects, and responds, continuously improving identify gaps and turn them into measurable gains in visibility and control.

The problem with point-in-time assessments



Not just an operational pain



SECURITY EXECUTIVES

"How do I prioritize what to fix, and how do I measure and prove the ROI of what we've invested in?"



SECURITY OPERATIONS

"How do I know my controls didn't regress since last week?"

"How do I efficiently engineer a detection for that new technique?"



OFFENSIVE SECURITY

"How do I scale my expertise and stop manually repeating the same simulations across every engagement?"

So how do we improve?

The **Gartner** Solution:



Adversarial Exposure Validation (AEV)

Gartner defines adversarial exposure validation (AEV) as technologies that deliver **consistent, continuous and automated** evidence of the feasibility of an attack. [...] AEV technologies provide automated execution of both simplified and/or extensible attack scenarios.



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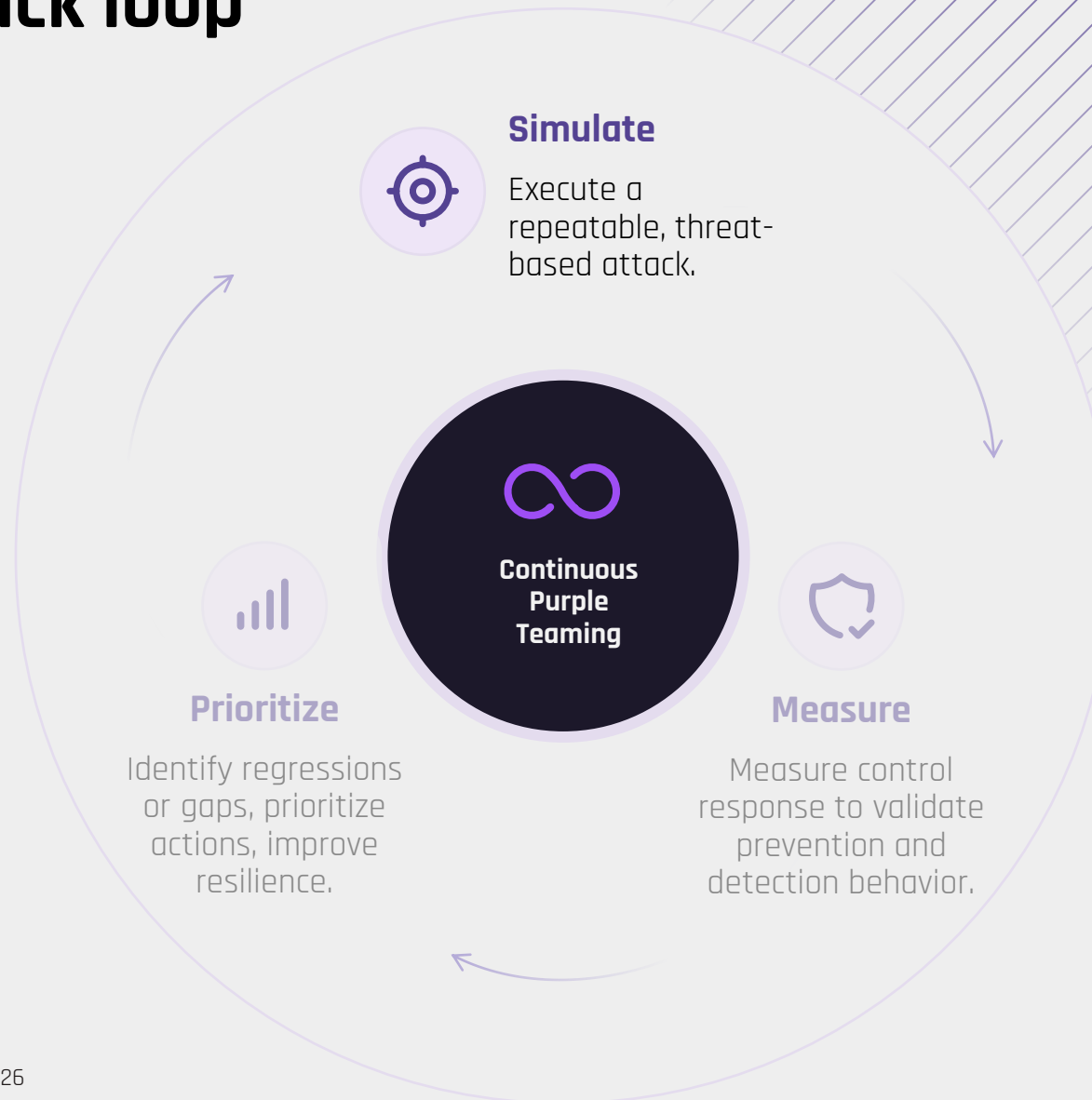
The Implementation:



Continuous Purple Teaming (CPT)

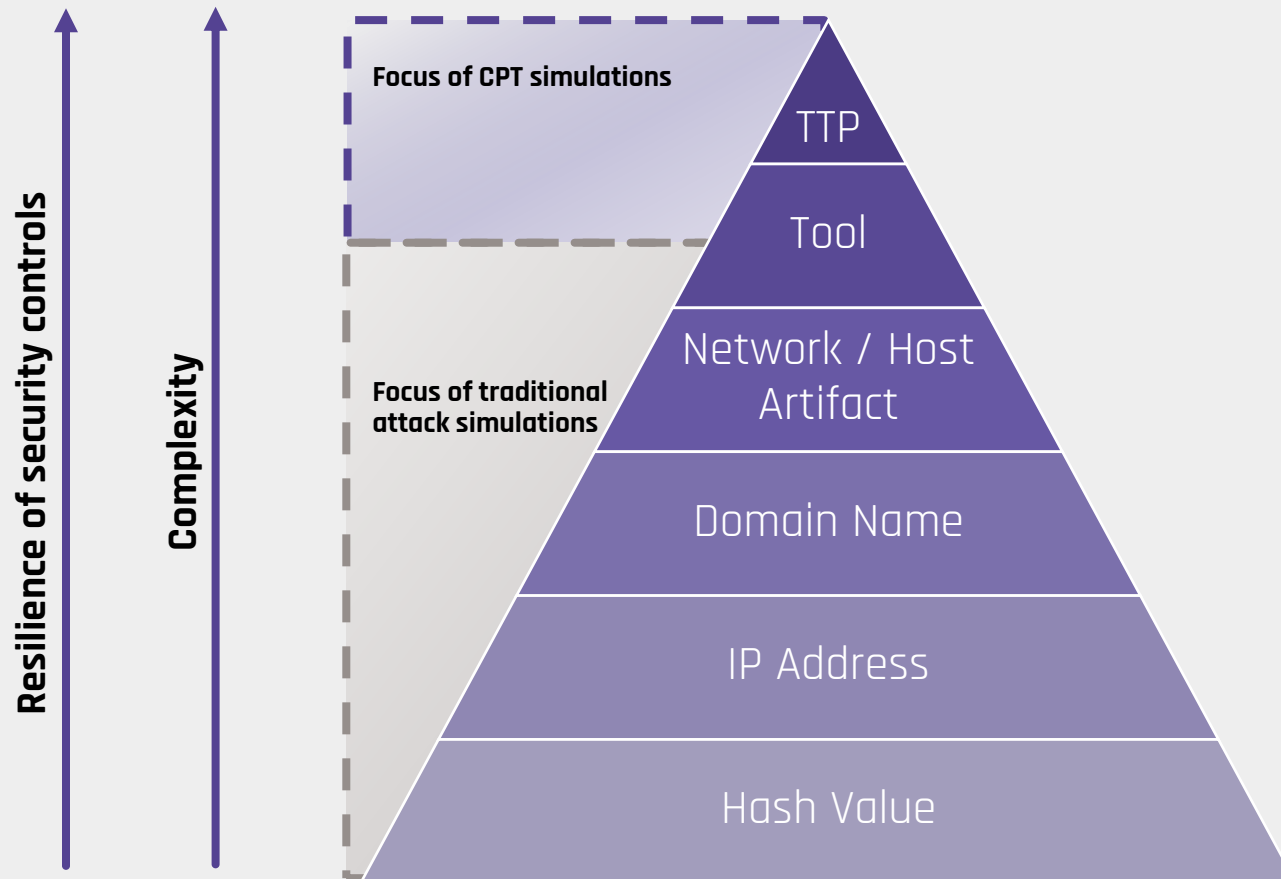
Continuous Purple Teaming is a form of AEV with an emphasis on delivering ongoing, high-value insights from an offensive security perspective to the blue team. CPT turns security simulations into a **continuous feedback loop** that drives measurable improvements across prevention and detection.

The CPT feedback loop



Simulating beyond the command line

Tool-agnostic simulations support the creation of resilient security controls





Three ways to simulate a technique



CRITICAL PATH

Simulate only the single critical path that a technique **must** execute (ideally in-memory).

e.g. Do not simulate SharpHound, but simulate the API/network calls



Reading tip “On Detection: Tactical to Functional” blog series by Jared Atkinson



Three ways to simulate a technique



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COVERAGE

Generate modular simulations across every variant of the technique to maximize coverage.

e.g. Process Injection (Kernel32 vs. NTDLL vs. EarlyBird vs. PoolParty)

ORDER OF PREFERENCE



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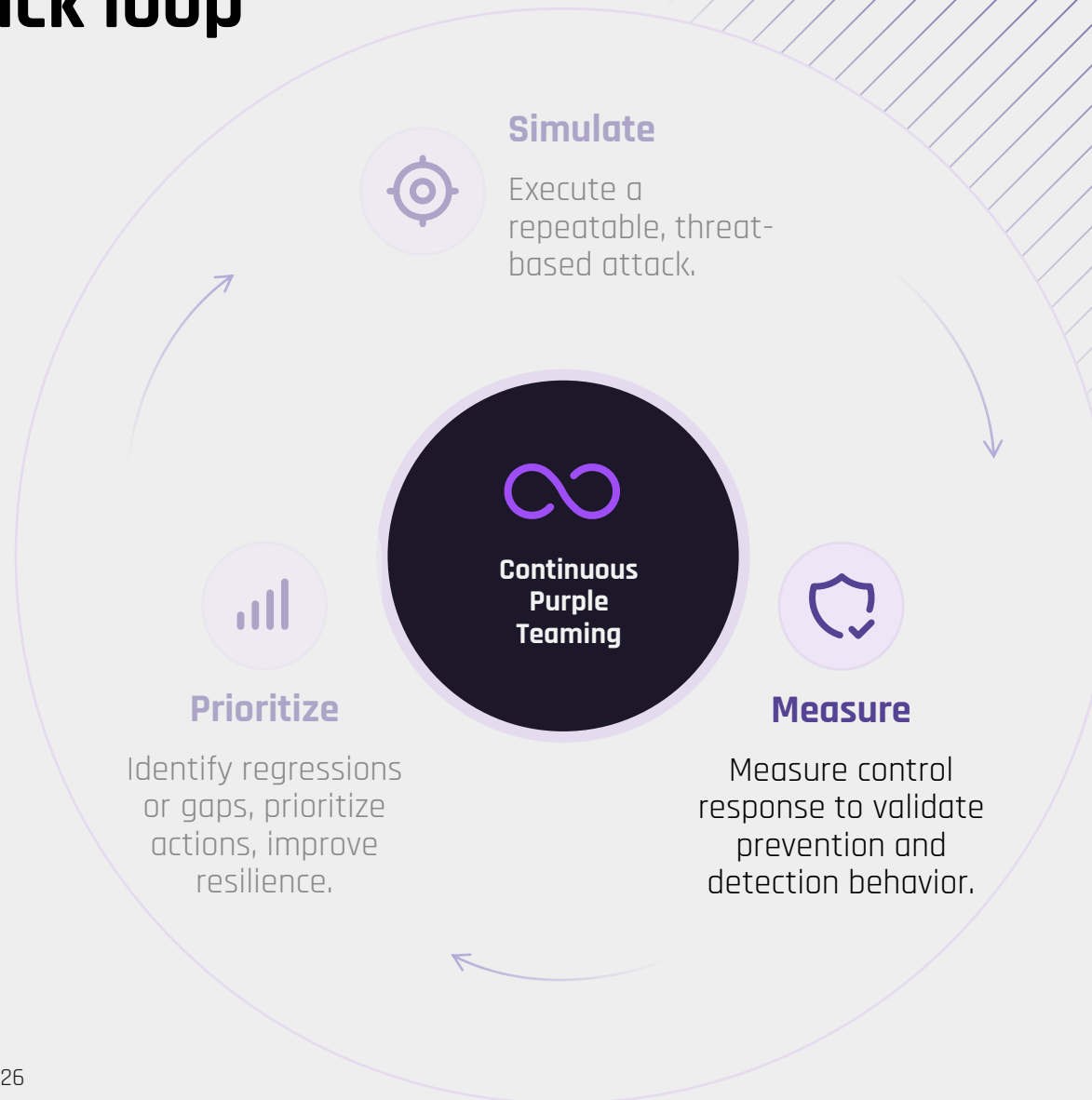
TOOL-BASED

Simulate based on specific tools, only if specifically specified by Threat Intelligence.

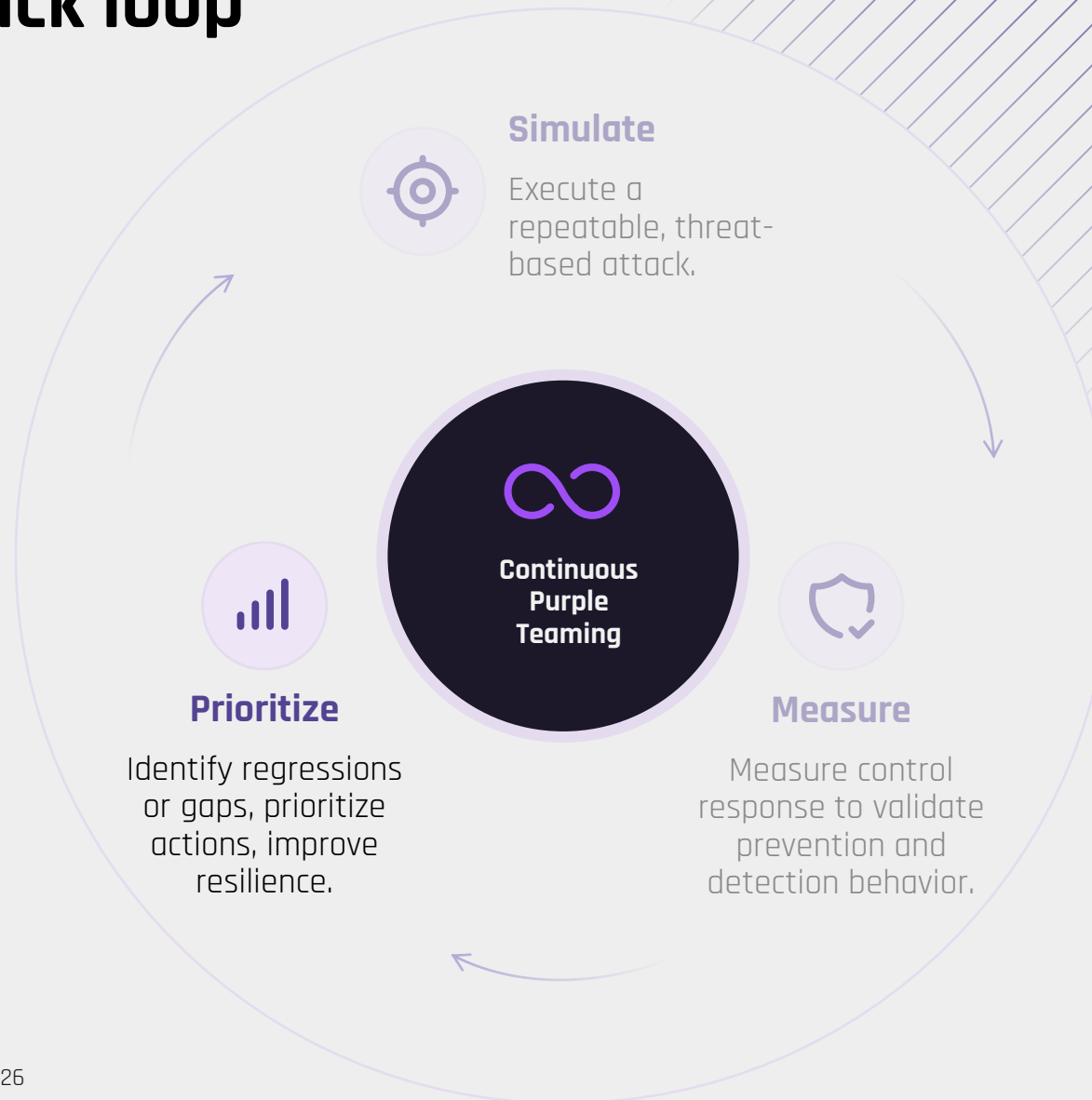
e.g. LOLBINs

ORDER OF PREFERENCE

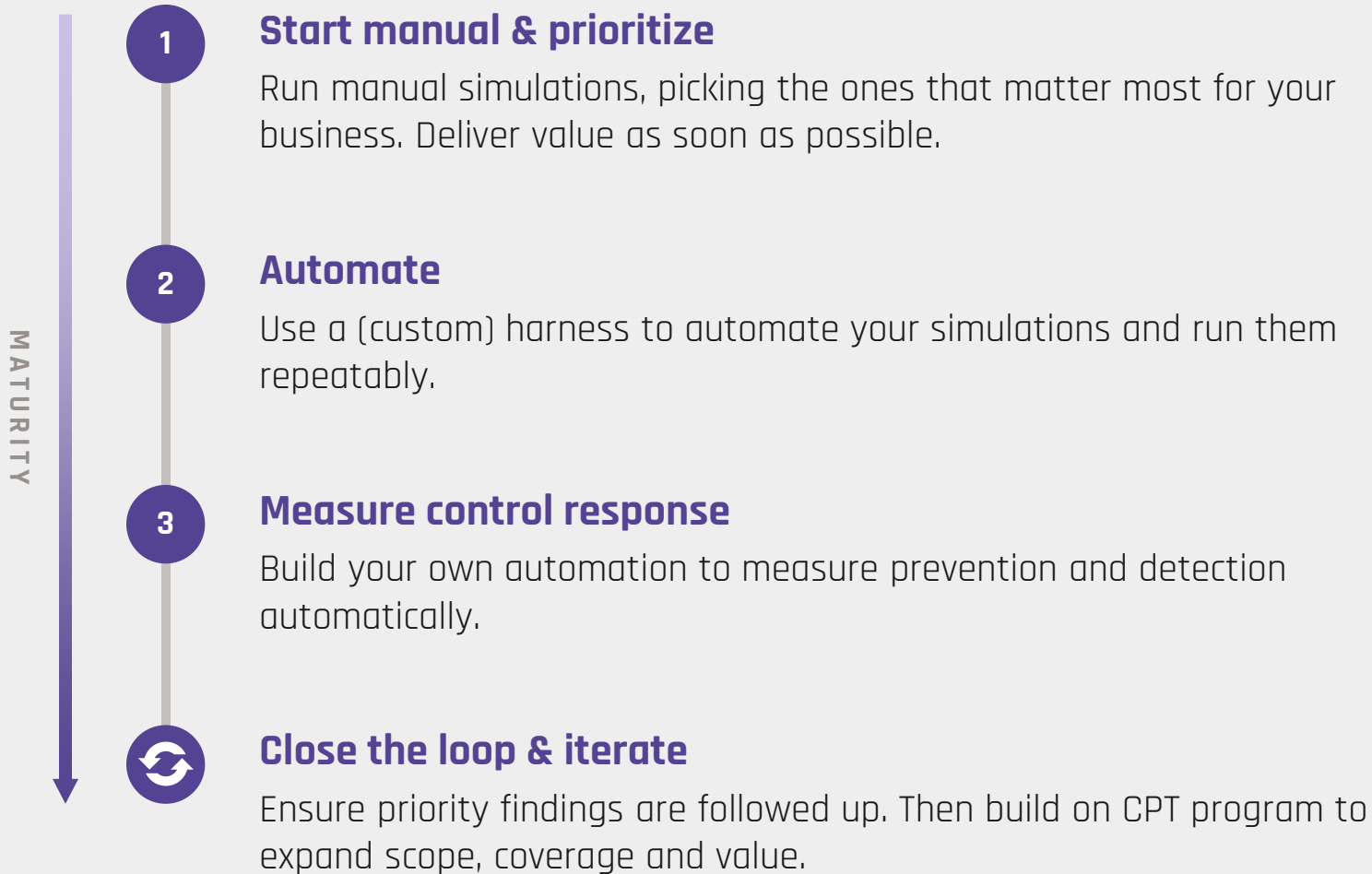
The CPT feedback loop



The CPT feedback loop



Getting started with CPT



Atomic Red Team
OPEN-SOURCE LIBRARY



MITRE Caldera
OPEN-SOURCE HARNESS



Takeaways

TAKEAWAY 1

Detections that are not regression tested can and will regress

TAKEAWAY 2

Simulate the technique behavior, not just the tool

TAKEAWAY 3

**Start with the 80/20:
Minimize time to value delivery**



Thank you!

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